

Hassan Ilyas Virk

Rawalpindi, PK | +92 303 543 2101 | hassanilyas.virk@gmail.com
linkedin.com/in/hassan-ilyas-virk | github.com/Hassan-Ilyas-Virk

Education

Bachelor of Science (BS) – Software Engineering

Aug 2022 – Jun 2026

National University of Computer and Emerging Sciences, FAST

Related Coursework: Web Development, Generative AI, Applied Artificial Intelligence, Design and Analysis of Algorithms, Probability and Statistics, Linear Algebra, Database Systems.

Work Experience

Freelance Software Developer

2025

- Built a MERN patient management system with JWT-based role access control for 500+ patients and cut scheduling latency by 40%.

Projects

ClothCraft AI (FYP) (FastAPI, React, PyTorch)

2026

- Built a generative fashion platform using Pix2Pix GANs for sketch-to-texture translation and fine-tuned Realistic Vision diffusion models for text-guided inpainting.
- Designed a FastAPI inference layer connecting the React frontend to the generative models, enabling real-time garment visualization.

Galaxy Generation CGANs (Python, PyTorch)

2025

- Trained a Physics-Aware CGAN to synthesize galaxy morphologies using the Galaxy Zoo 2 dataset.
- Added custom physics-informed loss functions to keep generated outputs consistent with astrophysical constraints.

End-to-End Encryption System (Node.js, Web Crypto API)

2025

- Built a zero-knowledge messaging platform using ECDH (P-384) for key exchange and AES-256-GCM for authenticated encryption.
- Added forward secrecy and ECDSA digital signatures; private keys stay in browser IndexedDB.

Proactive Agentic Survey System (Python, FastAPI, RAG)

2025

- Built an autonomous agent using RAG and vector databases to detect feedback opportunities and surface insights from unstructured data.
- Added a long-term memory module so the agent learns user preferences over time.

Skills

- Languages:** Python, JavaScript/TypeScript, Java, C++, SQL, HTML, CSS.
- Frameworks & Libraries:** React, Node.js, Express, React Native, FastAPI, PyTorch, RAG / Vector Databases.
- Tools:** Docker, Git, Figma, Jira, UML, JUnit.
- Domains:** Full-Stack Web, Deep Learning, Generative AI.

Honors & Certifications

- Certificate in Mobile Application Development** — University of Engineering and Technology (UET).
- Academic Excellence Scholarship** — Full tuition award for top ranking in Software Engineering.